



PERTH MODERN SCHOOL

Exceptional schooling. Exceptional students.

INDEPENDENT PUBLIC SCHOOL

Semester Two Examination, 2018

Question/Answer booklet

## MATHEMATICS METHODS UNIT 1 AND 2

### Section One: Calculator-free

Name: \_\_\_\_\_

Teacher's Name: \_\_\_\_\_

#### Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

#### Materials required/recommended for this section

##### *To be provided by the supervisor*

This Question/Answer booklet

Formula sheet

##### *To be provided by the candidate*

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,  
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

#### Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

| Question     | Mark |
|--------------|------|
| 1            |      |
| 2            |      |
| 3            |      |
| 4            |      |
| 5            |      |
| 6            |      |
|              |      |
| <b>TOTAL</b> |      |

**Structure of this paper**

| Section                         | Number of questions available | Number of questions to be answered | Working time (minutes) | Marks available | Percentage of examination |
|---------------------------------|-------------------------------|------------------------------------|------------------------|-----------------|---------------------------|
| Section One: Calculator-free    | 6                             | 6                                  | 50                     | 50              | 33                        |
| Section Two: Calculator-assumed | 9                             | 9                                  | 100                    | 100             | 67                        |
| <b>Total</b>                    |                               |                                    |                        |                 | 100                       |

**Instructions to candidates**

1. The rules for the conduct of the Western Australian Certificate of Education ATAR course examinations are detailed in the *Year 12 Information Handbook 2016*. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet.
3. You must be careful to confine your answers to the specific questions asked and to follow any instructions that are specific to a particular question.
4. Additional pages for the use of planning your answer to a question or continuing your answer to a question have been provided at the end of this Question/Answer booklet. If you use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.
5. **Show all your working clearly.** Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
6. It is recommended that you **do not use pencil**, except in diagrams.
7. The Formula sheet is **not** to be handed in with your Question/Answer booklet.

**Section One: Calculator-free**

**(50 Marks)**

This section has **six (6)** questions. Answer **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

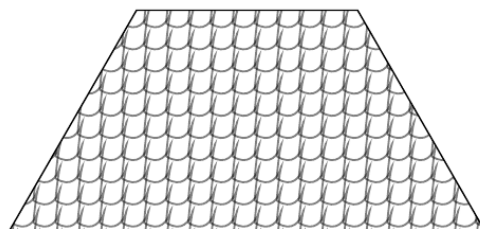
- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.

Working time: 50 minutes.

**Question 1**

**( 5 marks)**

The roof of a heritage building has roof tiles arranged neatly in horizontal rows. A sample diagram is shown below.



There are 28 roof tiles in the top row and each consecutive row has an extra 4 tiles than the row above it. If the bottom row has 96 tiles, show that there is a total of 1 116 tiles on the roof of the museum.

| Solution                                                                                                                                                                                                                                                                          |                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| $a_1 = 28$                                                                                                                                                                                                                                                                        | $96 = 28 + (n - 1)(4)$           |
| $a_n = 96$                                                                                                                                                                                                                                                                        | $96 = 24 + 4n$                   |
| $d = 4$                                                                                                                                                                                                                                                                           | $18 = n$                         |
|                                                                                                                                                                                                                                                                                   | $S_{18} = \frac{18}{2}(28 + 96)$ |
|                                                                                                                                                                                                                                                                                   | $= 1\,116$                       |
| $\therefore$ There are 18 rows of tiles, and a total of 1 116 tiles on the roof.                                                                                                                                                                                                  |                                  |
| <ul style="list-style-type: none"> <li>✓ identifying important info</li> <li>✓ setting up equation to find <math>n</math></li> <li>✓ the value of <math>n</math></li> <li>✓ setting up equation to find <math>S_n</math></li> <li>✓ showing that there are 1 116 tiles</li> </ul> |                                  |

## Question 2

( 5 marks)

Evaluate the expression below, giving your answer as a simplified fraction.

$$\left( 125^{\frac{1}{3}} \times 25^{\frac{1}{2}} + 16^{\frac{3}{4}} \times 64^{\frac{1}{3}} + \frac{1}{49^{\left(-\frac{1}{2}\right)}} \right)^{-\frac{2}{3}}$$

## Solution

$$\begin{aligned} &= \left( (5^3)^{\frac{1}{3}} \times (5^2)^{\frac{1}{2}} + (2^4)^{\frac{3}{4}} \times (2^6)^{\frac{1}{3}} + 7 \right)^{-\frac{2}{3}} \\ &= (5 \times 5 + 2^3 \times 2^2 + 7)^{-\frac{2}{3}} \\ &= (64)^{-\frac{2}{3}} \\ &= \left( 64^{\frac{1}{3}} \right)^{-2} \\ &= 4^{-2} \\ &= \frac{1}{16} \end{aligned}$$

✓✓ evaluating indices

✓applying BIMDAS

✓evaluated/simplified fraction

Question 3

(11 marks)

Given the function  $f(x) = 2x^2 - 6x - 10$ ,

(a) state  $f(x + h)$ . Do not simplify.

(2 marks)

| Solution                                |
|-----------------------------------------|
| $f(x + h) = 2(x + h)^2 - 6(x + h) - 10$ |
| ✓✓ (-✓ for every wrong term)            |

(b) set up and evaluate the difference quotient to find the derivative of the function.

(4 marks)

| Solution                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{2(x + h)^2 - 6(x + h) - 10 - (2x^2 - 6x - 10)}{h}$ $= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 - 6x - 6h - 10 - 2x^2 + 6x + 10}{h}$ $= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 6h}{h}$ $= \lim_{h \rightarrow 0} 4x + 2h - 6$ $= 4x - 6 \quad (\text{as } h \rightarrow 0)$ |
| ✓ setting up difference quotient with limit notation<br>✓ expanding $f(x + h)$<br>✓ cancelling terms<br>✓ derivative                                                                                                                                                                                                                                                 |

- (c) What is the instantaneous rate of change when  $x = -2$ ? (2 marks)

| Solution         |
|------------------|
| $f'(x) = 4x - 6$ |
| $f'(-2) = -14$   |
|                  |

- (d) Is the rate of change increasing or decreasing for  $x < -2$ ? Justify. (3 marks)

| Solution                                                                                |
|-----------------------------------------------------------------------------------------|
| Stationary point is when $f'(x) = 0$<br>$f'(x) = 4x - 6$<br>$0 = 2(2x - 3)$             |
| $\therefore$ Stationary point is when $x = 1.5$                                         |
| Taking a point to the left of $x = -2$ ,<br>$f'(-3) = -18$ .                            |
| Hence, the rate of change is decreasing for $x < -2$ .                                  |
| ✓ determining stationary points<br>✓ rate of change increasing<br>✓ valid justification |

Question 4

( 7 marks)

Given the equation  $2^{x+1} + 2^{3-x} = 17$ ,  $x \in \mathbb{R}$ ,

- (a) use a suitable substitution to rewrite the equation above as a quadratic equation, (3 marks)

| Solution                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------|
| $2^{x+1} + 2^{3-x} = 17$ $2^x \times 2 + 2^3 \times 2^{-x} = 17 \quad \text{let: } 2^x = y$ $2y + \frac{8}{y} = 17$ $2y^2 - 17y + 8 = 0$ |
| <p>✓ using appropriate substitution</p> <p>✓✓ quadratic equation (-✓ for every wrong term)</p>                                           |

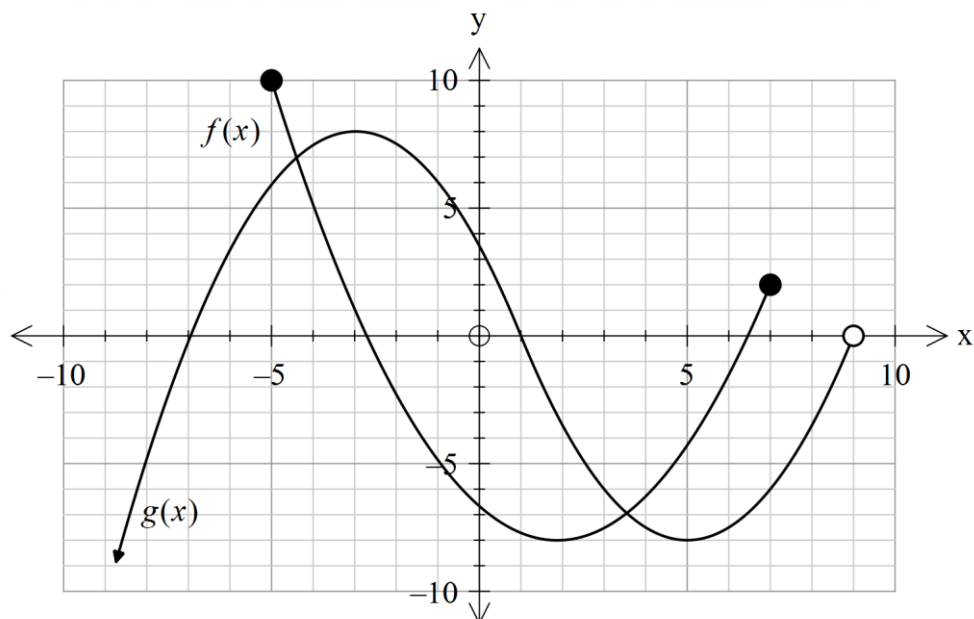
- (b) hence, determine the solution(s) to the exponential equation. (4 marks)

| Solution                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $2y^2 - 17y + 8 = 0$ $(2y - 1)(y - 8) = 0$ $\therefore y = \frac{1}{2} \text{ or } 8.$ $2^x = \frac{1}{2} \quad \rightarrow x = -1$ $2^x = 8 \quad \rightarrow x = 3$ |
| <p>✓ factorising quadratic</p> <p>✓ y values</p> <p>✓✓ for solutions to the exponential equation</p>                                                                  |

Question 5

(11 marks)

The graphs of the functions  $f(x)$  and  $g(x)$  are shown below.



- (a) State the domain of  $g(x)$ . (1 mark)

| Solution                      |
|-------------------------------|
| $\{x \in \mathbb{R}: x < 9\}$ |
|                               |

- (b) State the range of  $f(x)$ . (1 mark)

| Solution                                        |
|-------------------------------------------------|
| $\{f(x) \in \mathbb{R}: -8 \leq f(x) \leq 10\}$ |
|                                                 |

- (c) Which function(s), if any, have a point of inflection? (1 mark)

| Solution |
|----------|
| $g(x)$   |
|          |

- (d) Describe how the graph of  $y = -g(x)$  may be obtained from the graph of  $y = g(x)$  and hence state the coordinates of the turning point(s) of the graph  $y = -g(x)$ . (3 marks)

| Solution                                                                        |
|---------------------------------------------------------------------------------|
| Reflection about the x axis.                                                    |
| Turning points: $(-3, 8) \rightarrow (-3, -8)$ and $(5, -8) \rightarrow (5, 8)$ |
| ✓ identifying transformation                                                    |
| ✓✓ correct turning points                                                       |

- (e) Describe how the graph of  $y = f(x + 1)$  may be obtained from the graph of  $y = f(x)$  and hence state the coordinates of the endpoints of the graph  $y = f(x + 1)$ . (3 marks)

| Solution                                                                    |
|-----------------------------------------------------------------------------|
| Horizontal translation of 1 unit to the left.                               |
| End points: $(-5, 10) \rightarrow (-6, 10)$ and $(7, 2) \rightarrow (6, 2)$ |
| ✓ identifying transformation                                                |
| ✓✓ correct end points                                                       |

- (f) State the approximate solution(s) to the equation  $f(x) = g(x)$ . (2 marks)

| Solution                                                           |
|--------------------------------------------------------------------|
| $\left(4\frac{1}{3}, 7\right)$ and $\left(3\frac{1}{2}, -7\right)$ |
|                                                                    |

Question 6

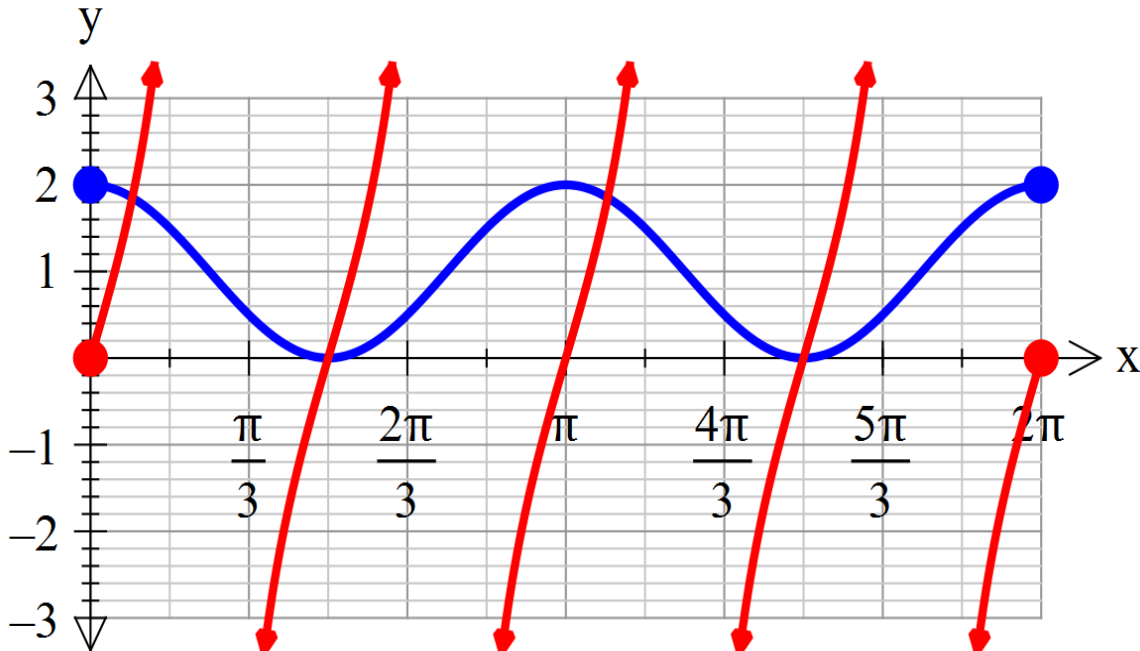
(12 marks)

Given the following equations:

$$y_1 = 2 \cos 2x, \quad y_2 = 3 \tan 2x, \quad 0 \leq x \leq 2\pi.$$

- (a) Sketch on a single set of axes the graph of  $y_1$  and  $y_2$ .

(4 marks)



| Solution           |  |
|--------------------|--|
| See graph above.   |  |
| For each function: |  |
| ✓ shape            |  |
| ✓ accuracy         |  |

- (b) Show that the coordinates of the points of intersection between the graphs of  $y_1$  and  $y_2$  are solutions of the equation  $2 \sin^2 2x + 3 \sin 2x - 2 = 0$ .

(4 marks)

| Solution                                |  |
|-----------------------------------------|--|
| $2 \cos 2x = 3 \tan 2x$                 |  |
| $2 \cos 2x = \frac{3 \sin 2x}{\cos 2x}$ |  |
| $2 \cos^2 2x = 3 \sin 2x$               |  |
| $2 (1 - \sin^2 2x) = 3 \sin 2x$         |  |
| $2 \sin^2 2x + 3 \sin 2x - 2 = 0$       |  |
| ✓ rewriting tan in terms of sin and cos |  |
| ✓ multiplying both sides by cos         |  |
| ✓ using Pythagorean identity            |  |
| ✓ required equation                     |  |

- (c) Hence find the  $x$  coordinates of the points of intersection between the graphs of  $y_1$  and  $y_2$  over the domain  $[0, 2\pi]$ . (4 marks)

| Solution                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $let \sin 2x = y$<br>$2y^2 + 3y - 2 = 0$<br>$(y + 2)(2y - 1) = 0 \quad \rightarrow \quad y = -2 \quad or \quad \frac{1}{2}$<br>$\rightarrow \sin 2x = -2 \quad NO \ SOLUTIONS$<br>$\rightarrow \sin 2x = \frac{1}{2}$<br>$2x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$<br>$x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$ |
| <p>✓ factorizing quadratic and finding two solutions</p> <p>✓ indicating that there are no solutions for <math>\sin 2x = -2</math></p> <p>✓ ✓ finding the 4 solutions (-✓ for every incorrect answer)</p>                                                                                                                                                                    |

**Additional working space**

Question number: \_\_\_\_\_

**Additional working space**

Question number: \_\_\_\_\_

**Additional working space**

Question number: \_\_\_\_\_

**Additional working space**

Question number: \_\_\_\_\_

